

Signed Schur Cancellation in Cross-Prime Gram Pairs: a Finite Coherence-to-Cancellation Atlas

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Abstract

The preceding physical Gram experiments found a late-shell regime with strong positive cross-prime coherence and an isolated early sign-flip regime. This paper asks what the corresponding two-component cancellation directions actually cost. For two nonzero physical prime-component vectors g_i, g_j , with $d_k = \|g_k\|^2$ and $G_{ij} = \langle g_i, g_j \rangle$, we prove the exact projection identity

$$\min_{\rho \in \mathbb{R}} \frac{\|g_i - \rho g_j\|^2}{d_i} = 1 - \Gamma_{ij}, \quad \Gamma_{ij} = \frac{G_{ij}^2}{d_i d_j},$$

and the normalized signed two-vector Rayleigh minimum $1 - \sqrt{\Gamma_{ij}}$. The sign of the optimal projection coefficient then gives an explicit distinction between opposite-sign cancellation for a positive cross term and same-sign cancellation for a negative one. An exact-rational replay of the 18-row TPC-289 grid evaluates all 1,380 pairs: 1,377 are positive and 3 are negative, while 1,074, 852, and 477 pairs have Schur residual at most $1/2$, $1/4$, and $1/10$. The best tested pair has $\Gamma \approx 0.984876$ and residual ≈ 0.015124 . These results quantify a genuine pairwise cancellation mechanism, but do not establish a multi-prime signed null direction, an arithmetic L^2 estimate, or a twin-prime theorem.

1 Position on the twin-prime route

This release stays inside the same literal prime-shell dynamical family. TPC-287 separated the physical deleted-diagonal output into prime components, and TPC-288 showed that retaining those full output vectors produces a positive-semidefinite source-output Gram object whose finite energy can increase even when a scalar readout is small [4, 3]. TPC-289 made the cross terms explicit: most tested pairs were positive and a late block had large normalized coherence, but three negative pairs appeared at the early crossover row [2]. TPC-290 then proved that diffuse nonnegative shell weights cannot cross an all-positive coherence wall [1].

The next natural question is therefore narrower than “find a good weight”: when a pair is coherent, what is the exact best two-prime cancellation, and does its coefficient sign agree with the available shell mechanism? The answer is a clean local compiler. It also identifies the next bridge problem: assemble many such pair directions while preserving the literal prime-source arithmetic and the required normalization.

2 Frozen physical model

Let $I = I_N$ be the finite integer interval and let β be the frozen rational source vector inherited from the parent certificate. For an odd prime q , height H , and kernel exponent s , write

$$K_{H,s}(h) = \frac{H^{2s}}{(H^2 + h^2)^s}, \quad B_q(u, t) = \mathbf{1}_{q \nmid u} \mathbf{1}_{q \nmid t} \left(\mathbf{1}_{u \equiv t \pmod{q}} - \frac{1}{q-1} \right).$$

The physical deleted-diagonal component is the vector $g_q = A_q \beta$, where

$$(A_q)_{u,t} = q K_{H,s}(u-t) \mathbf{1}_{u \neq t} B_q(u, t).$$

For a shell S of primes, define

$$d_q = \|g_q\|^2, \quad G_{q,r} = \langle g_q, g_r \rangle, \quad \Gamma_{q,r} = \frac{G_{q,r}^2}{d_q d_r}.$$

The certificate verifies $d_q > 0$ on every declared row. By Cauchy-Schwarz, $0 \leq \Gamma_{q,r} \leq 1$; the square deliberately keeps the magnitude of coherence separate from its sign.

3 Exact two-prime cancellation compiler

We orient a pair as “target minus coefficient times reference”. Thus the quantity below is $\|g_i - \rho g_j\|^2$, with j the reference component.

Lemma 1 (Schur projection identity). *For nonzero real vectors g_i, g_j , the unique minimizer of $\|g_i - \rho g_j\|^2$ is*

$$\rho_* = \frac{G_{ij}}{d_j},$$

and

$$\min_{\rho \in \mathbb{R}} \frac{\|g_i - \rho g_j\|^2}{d_i} = 1 - \frac{G_{ij}^2}{d_i d_j} = 1 - \Gamma_{ij}. \quad (1)$$

Proof. Expanding the square gives

$$\|g_i - \rho g_j\|^2 = d_i - 2\rho G_{ij} + \rho^2 d_j.$$

Since $d_j > 0$, completing the square gives the unique minimizer $\rho_* = G_{ij}/d_j$. Substitution yields $d_i - G_{ij}^2/d_j = d_i(1 - \Gamma_{ij})$. $\square$

Proposition 1 (Signed two-vector Rayleigh minimum). *For real $(a, b) \neq (0, 0)$,*

$$\inf_{(a,b) \neq (0,0)} \frac{\|ag_i + bg_j\|^2}{a^2 d_i + b^2 d_j} = 1 - \sqrt{\Gamma_{ij}}. \quad (2)$$

Proof. Conjugating the two-by-two Gram matrix by $\text{diag}(d_i^{-1/2}, d_j^{-1/2})$ gives

$$\begin{pmatrix} 1 & c \\ c & 1 \end{pmatrix}, \quad c = \frac{G_{ij}}{\sqrt{d_i d_j}}.$$

Its eigenvalues are $1 + c$ and $1 - c$, so the smaller one is $1 - |c| = 1 - \sqrt{\Gamma_{ij}}$. $\square$

Corollary 1 (Sign cost). *In the orientation of Lemma 1, $\text{sign}(\rho_*) = \text{sign}(G_{ij})$. A positive cross term therefore uses opposite signs in $g_i - \rho_* g_j$, whereas a negative cross term uses same-sign coefficients after the subtraction is expanded.*

Proof. The denominator d_j is positive. The statement follows directly from the formula for ρ_* . Equivalently, the lower eigenvector in (2) is $(1, -1)$ when $c > 0$ and $(1, 1)$ when $c < 0$. $\square$

Remark 1. Equation (1) is a two-dimensional statement. It does not say that the individually optimal directions for different pairs are compatible, nor that their coefficients are produced by the literal prime source. This compatibility distinction is the central firewall of the present paper.

4 Finite protocol

The certificate replays eight $s = 2$ growth anchors, four $s = 1$ crossover rows, and six $s = 2$ source-control rows, for 18 rows in total. Each row uses the same interval, rational source, kernel, deleted diagonal, and prime shell as TPC-289. Every unordered pair in every shell is evaluated with exact rational arithmetic. The producer records the cross term, its sign, Γ , the exact residual $1 - \Gamma$, the projection coefficient, and the sign-cost label. The independent checker reconstructs the physical outputs in reverse column-first order. A ten-mutation hostile audit tests theorem fields, threshold counts, sign labels, row deletion, and provenance locks.

The finite thresholds are $1/2$, $1/4$, and $1/10$ for the residual, and $9/25$ and $3/4$ for squared coherence. These are audit thresholds, not claimed universal constants. The parent TPC-290 code, result, and frozen TPC-268 engine are SHA-256 locked by the producer.

5 Results

5.1 Global census

The 18 rows contain 1,380 unordered pairs. All exact Schur residuals are nonnegative, as required by the Gram determinant inequality. The sign and threshold census is:

Table 1: Finite pairwise census on the declared 18-row grid.

quantity	count
rows	18
unordered pairs	1,380
positive cross terms	1,377
negative cross terms	3
zero cross terms	0
$1 - \Gamma \leq 1/2$	1,074
$1 - \Gamma \leq 1/4$	852
$1 - \Gamma \leq 1/10$	477
$\Gamma \geq 9/25$	1,189
$\Gamma \geq 3/4$	852

Thus high coherence is common in this finite probe, but its sign is almost always positive. The sign census matters because the high-coherence direction for a positive pair is signed subtraction, while a nonnegative shell combination can only directly exploit the exceptional negative pairs.

5.2 Representative rows

Table 2 reports the most coherent pair on selected rows. The last column gives the sign of the cross term, which is also the sign of the optimal projection coefficient in the chosen orientation.

Table 2: Selected best-pair records. Decimal values are display-only; the certificate stores exact fractions.

axis	N	H	Q	s	pair	Γ	$1 - \Gamma$	sign
growth	128	24	9	2	(13,17)	0.364275	0.635725	+
growth	192	32	16	2	(17,23)	0.505386	0.494614	+
growth	384	50	40	2	(71,79)	0.758148	0.241852	+
growth	512	58	70	2	(131,137)	0.960038	0.039962	+
growth	512	58	90	2	(173,179)	0.984876	0.015124	+
crossover	256	38	27	1	(37,47)	0.236671	0.763329	+
control	384	52	70	2	(137,139)	0.974785	0.025215	+

The global best is the late growth row $(N, H, Q, z, s) = (512, 58, 90, 5, 2)$, with pair (173, 179) and residual approximately 0.0151239493. The largest coherence is therefore a real finite cancellation opportunity in the two-dimensional span, but it is not evidence of a prime gap or a shell-wide null vector.

5.3 The exceptional sign-flip row

The only negative pairs occur at the crossover row $(N, H, Q, z, s) = (256, 38, 27, 5, 1)$. They are

$$(29, 53), \quad (31, 53), \quad (41, 53).$$

For equal two-prime support, the physical ratio is

$$\frac{\|g_i + g_j\|^2}{d_i + d_j} = 1 + \frac{2G_{ij}}{d_i + d_j}.$$

Consequently these three negative pairs are precisely the finite same-sign subunit witnesses seen by TPC-290. They do not occur in the late positive block, where the low-residual directions require signed subtraction. The apparent “escape” is therefore sparse and localized; it does not yet provide a diffuse or growing-shell mechanism.

6 Interpretation and claim firewall

The results make the local geometry of the map more precise:

1. High cross-prime coherence converts exactly into a small two-prime Schur residual. This is a reusable analytic bridge from the Gram phase diagram to a cancellation direction.
2. The sign of the cross term is not cosmetic. In the dominant positive block, cancellation requires signed coefficients; the three same-sign opportunities are confined to one early row.
3. Pairwise optimality does not compose automatically. A multi-prime construction must control the compatibility of projection directions, coefficient signs, and the literal source normalization.

Accordingly, the strongest exact claim is the two-prime projection and Rayleigh compiler. The strongest finite numerical claim is the complete 1,380-pair coherence-to-cancellation atlas. The following remain open: multi-prime signed reassembly, a growing-shell theorem, a source-native arithmetic L^2 estimate, fixed-power credit, and the full Gate-B bridge. No Route-A or Route-B evaluator pass is claimed here because the Session-named evaluator files are absent from this checkout; the local proof, exact certificate, independent replay, stress audit, and Bridge-B checker are the fail-closed validation package.

7 Reproducibility and conclusion

From the project directory, the producer and independent replay can be run with:

```
export PYTHONDONTWRITEBYTECODE=1
python -B code/tpc291_signed_schur_cancellation_certificate.py --check
python -B experiments/tpc291_independent_checker.py
python -B experiments/tpc291_schur_stress.py
```

The canonical result is `results/tpc291_certificate.json`; normal and optimized runs are required to have empty standard error and byte-identical output.

TPC-291 closes one local uncertainty on the broad twin-prime map. It shows that the late coherent island is not missing a two-prime cancellation formula; the formula is exact. The unresolved bridge is higher-dimensional: whether compatible signed directions can be assembled from the actual prime-shell source while paying the physical L^2 and arithmetic budgets. That is the next meaningful question, rather than another unstructured weight scan.

References

- [1] Liang Wang. Adaptive shell weighting and the nonnegative coherence wall, 2026. TPC-290 project release in the same research repository.
- [2] Liang Wang. Cross-prime gram coherence in a literal twin-prime dynamics, 2026. TPC-289 project release in the same research repository.
- [3] Liang Wang. Growing-shell gram obstruction in a literal twin-prime dynamics, 2026. TPC-288 project release in the same research repository.
- [4] Liang Wang. Prime-shell cancellation depth in a literal twin-prime dynamics, 2026. TPC-287 project release in the same research repository.